

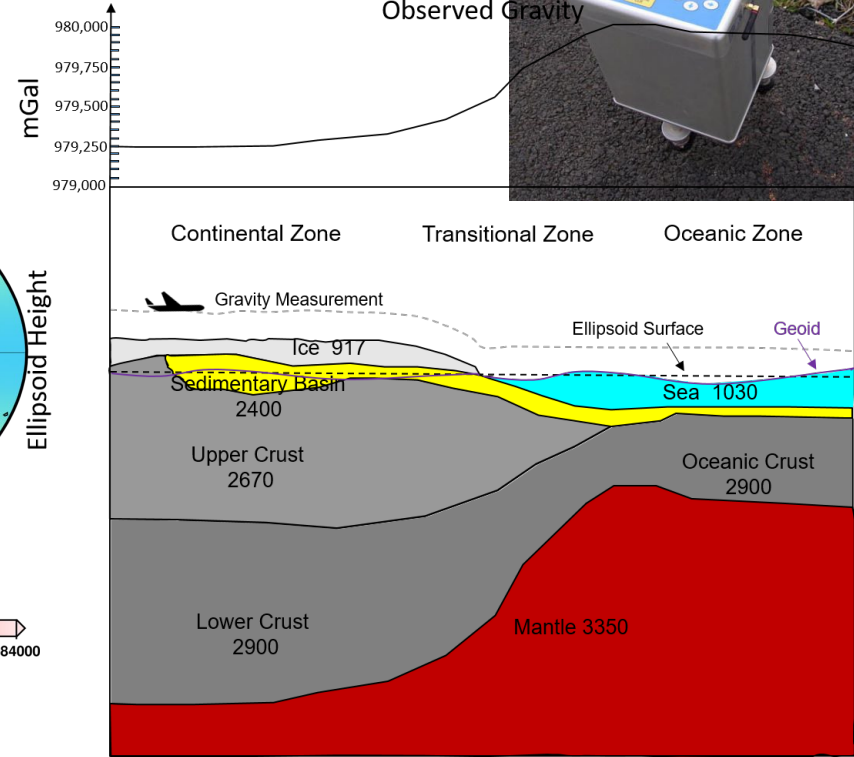
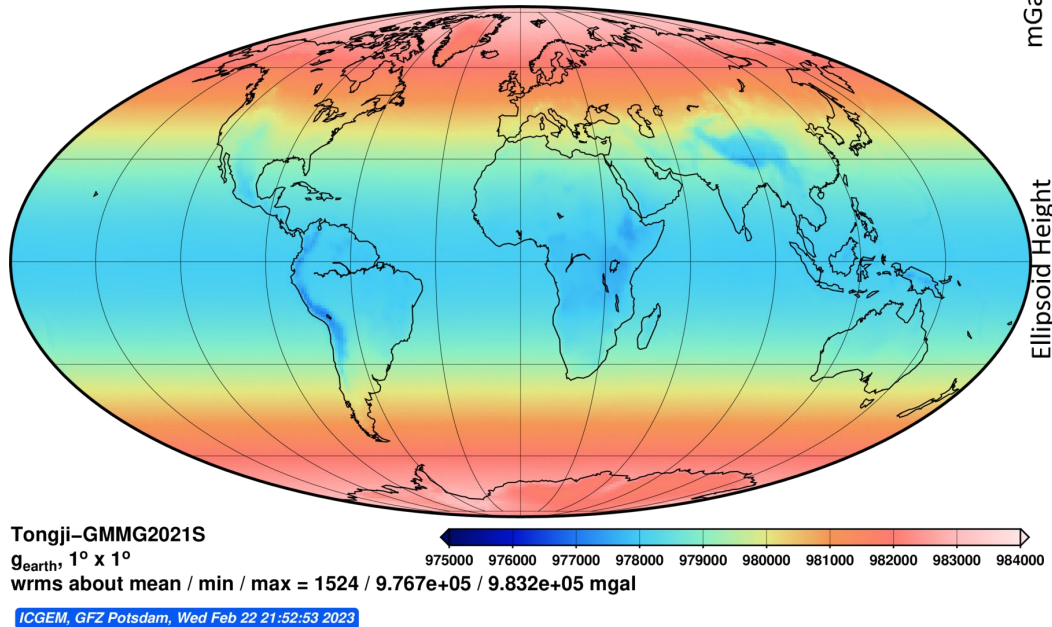


Gravity processing and modelling – CAGE 2026

Lu Li

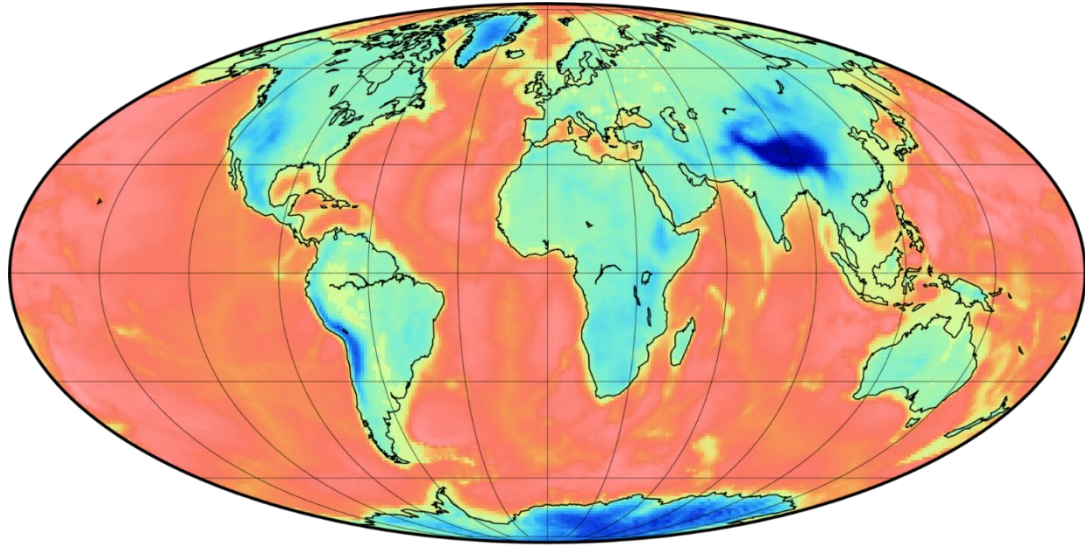


What do we measure?



Earth Gravity at surface topography

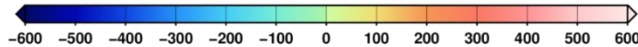
What do me want?



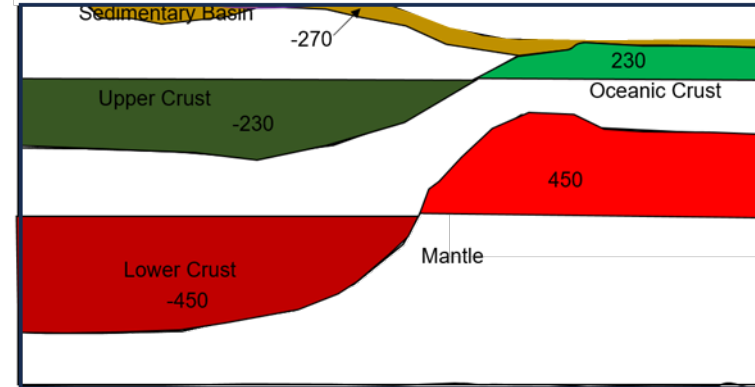
Tongji-GMMG2021S

$\Delta g_B, 1^\circ \times 1^\circ$

wrms about mean / min / max = 185.3 / -586.5 / 454.2 mgal

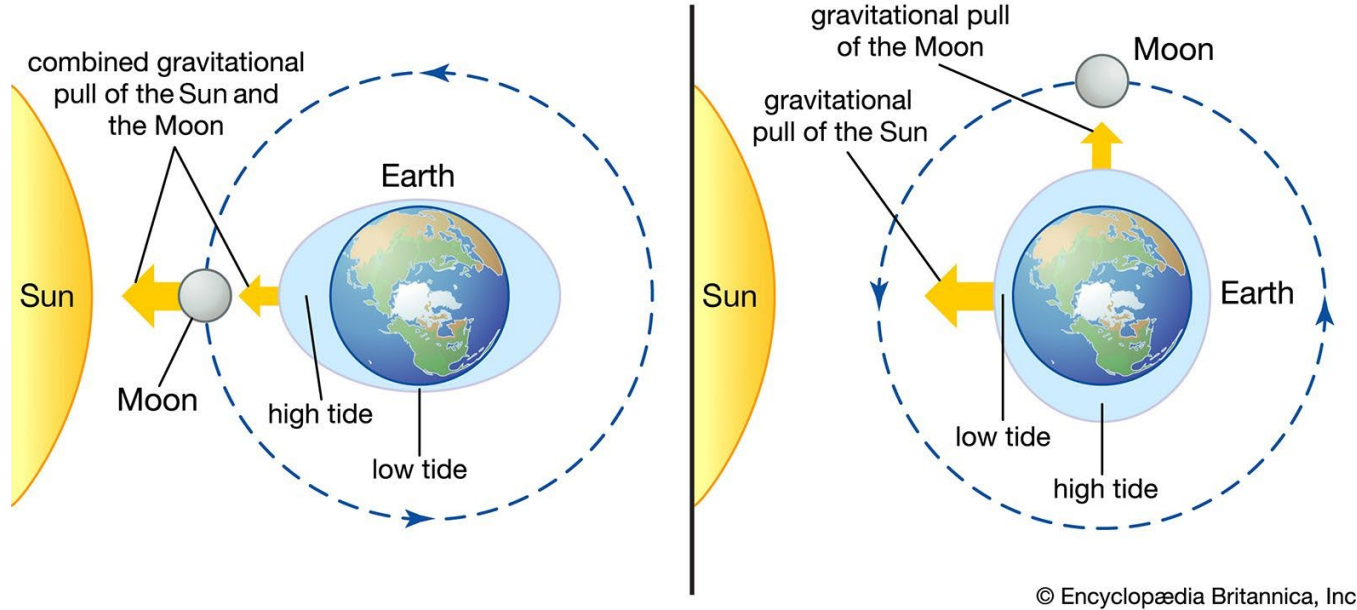


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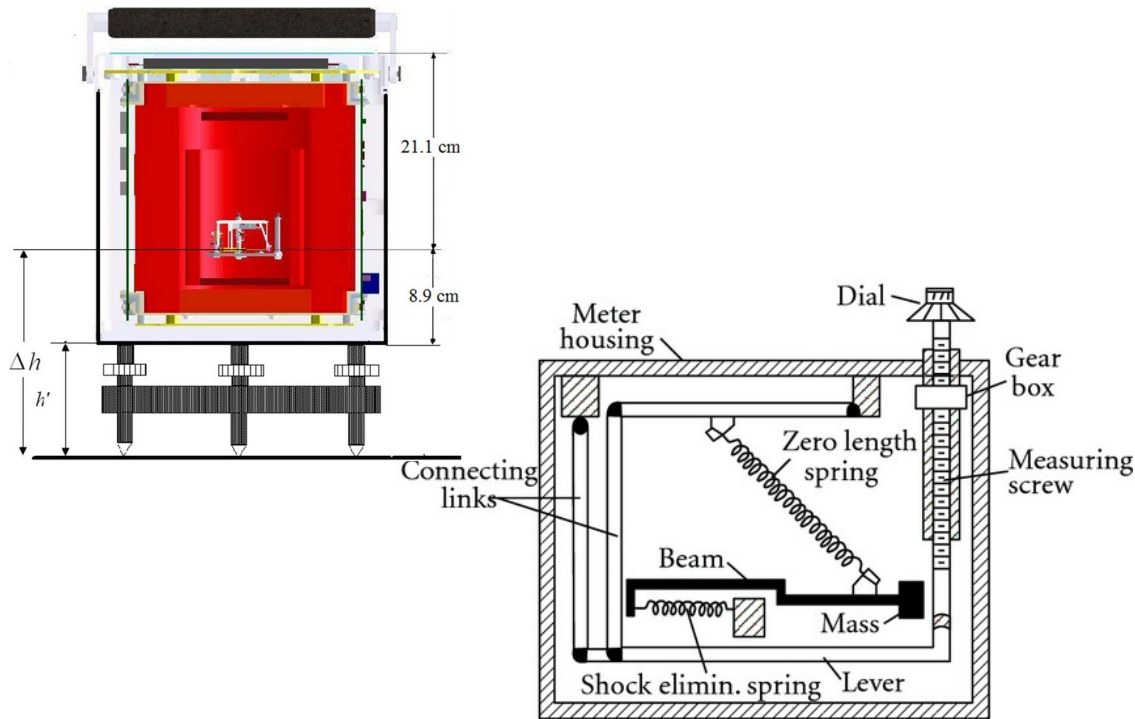
Bouguer Gravity

We need few steps to get here..

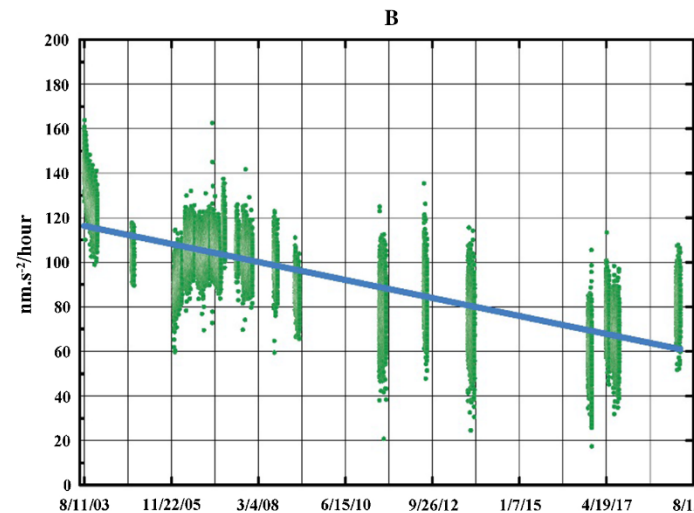
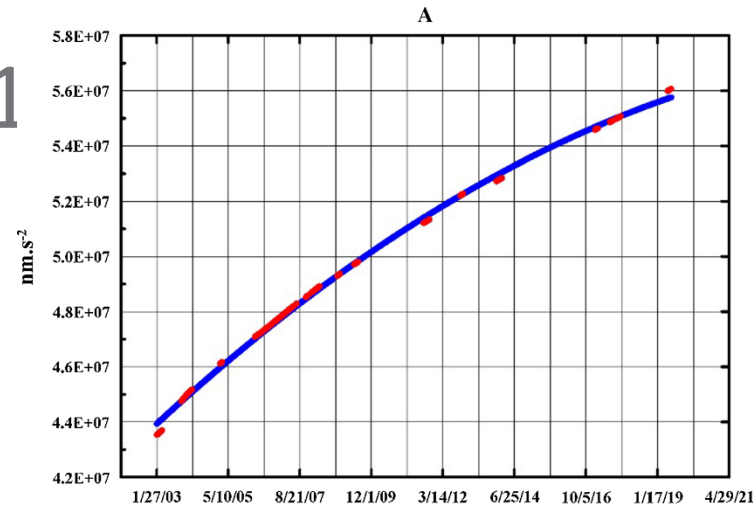


Tide correction

Drift correction – Notebook 1

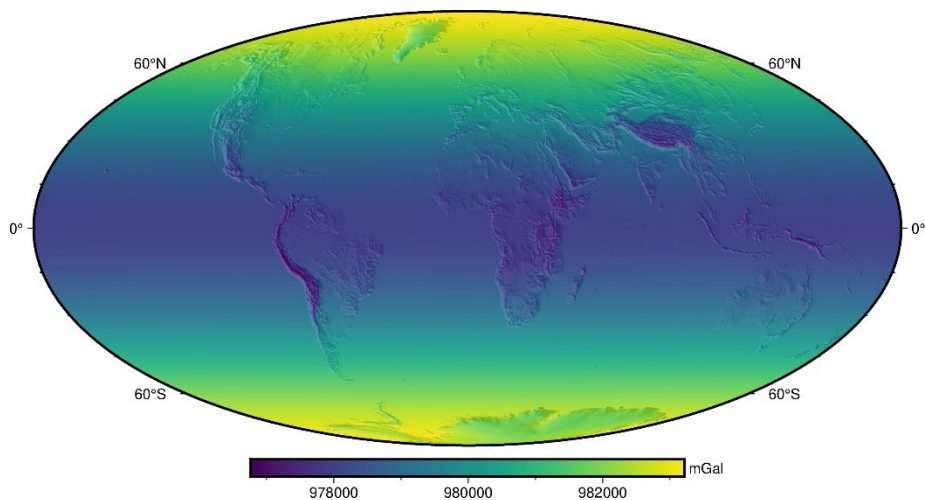
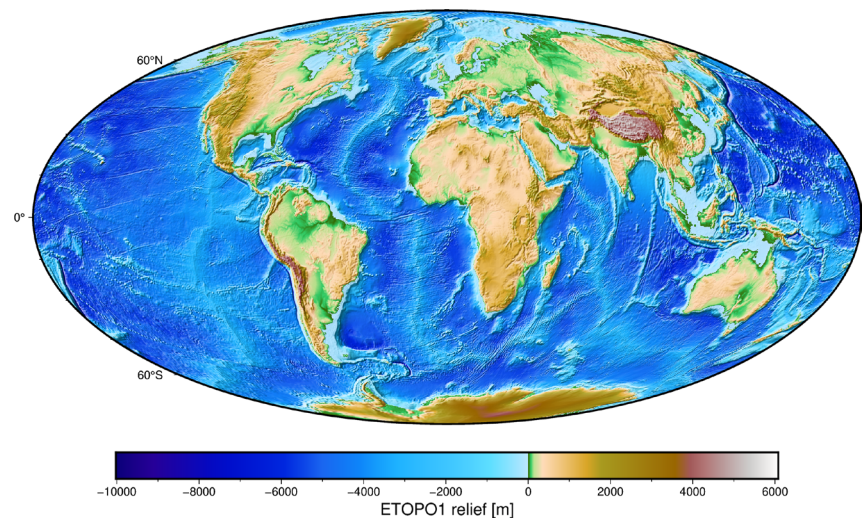


Drift correction



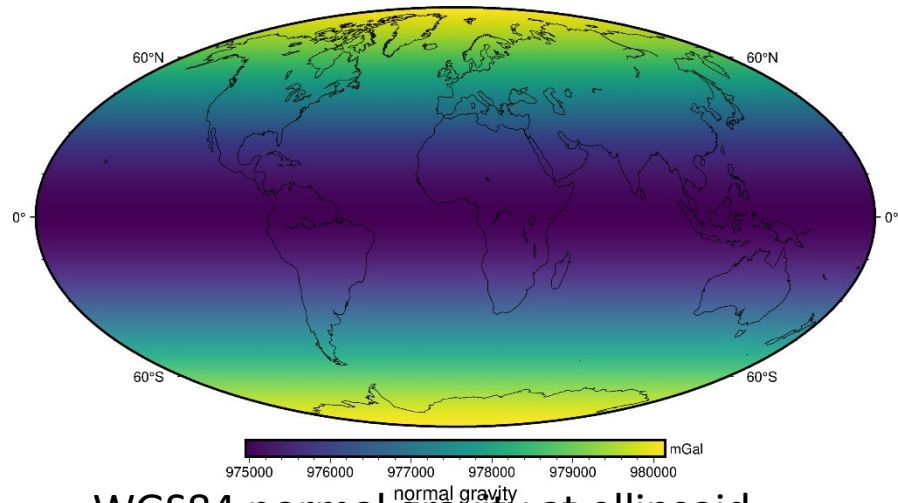
Correct 'reference earth'

OFFICIAL



WGS84 normal gravity at earth surface

OFFICIAL

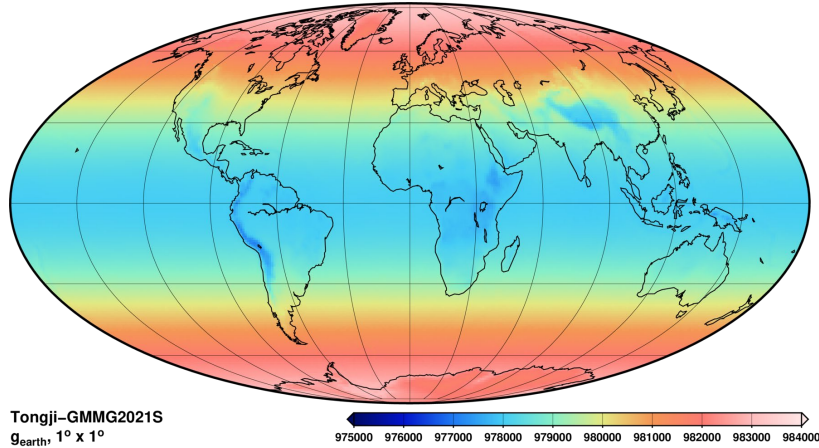


WGS84 normal gravity at ellipsoid

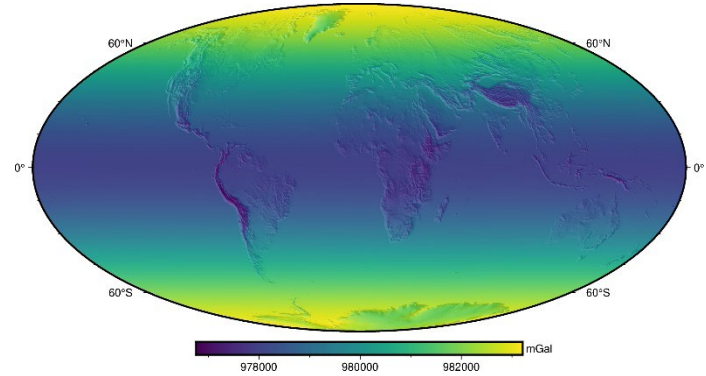
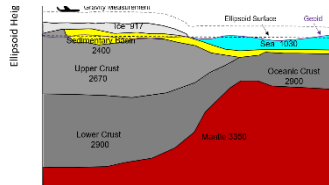


Correct 'reference earth'

- Earth Gravity field

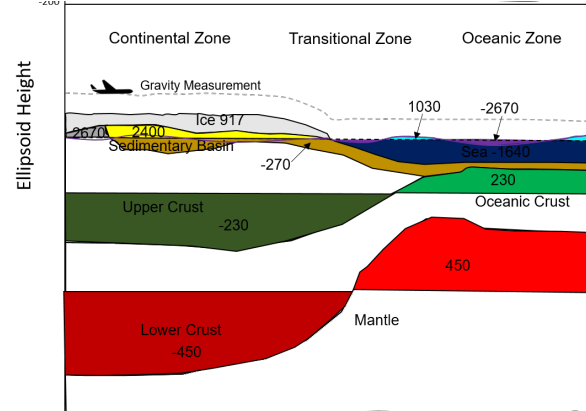


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Normal gravity (latitude variation + height variation)

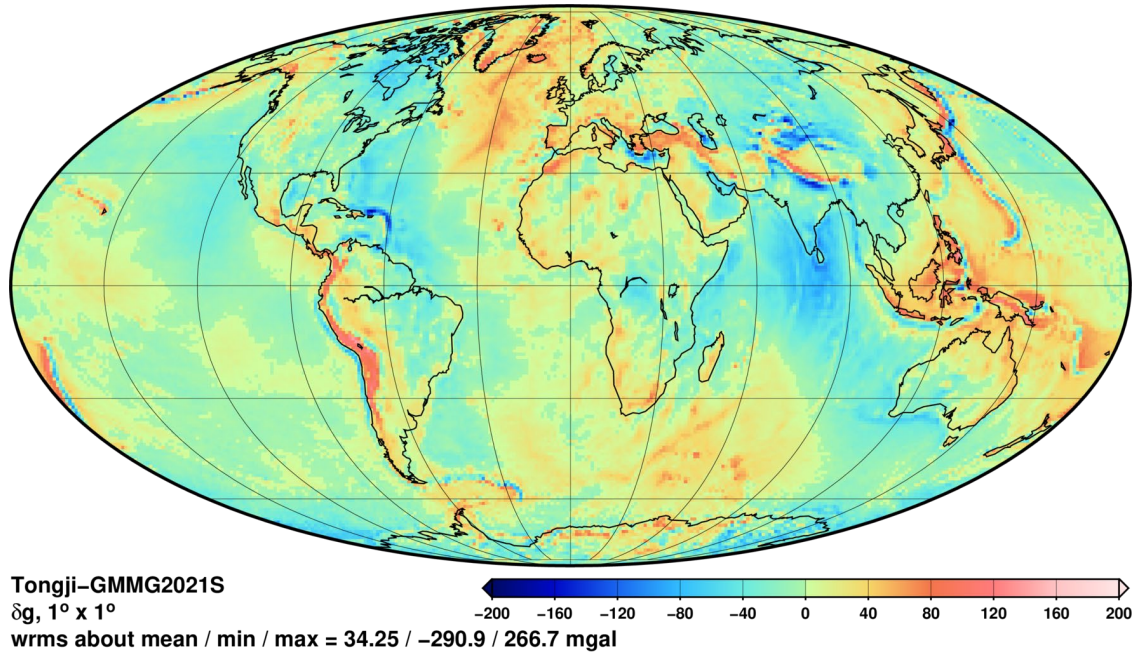
+



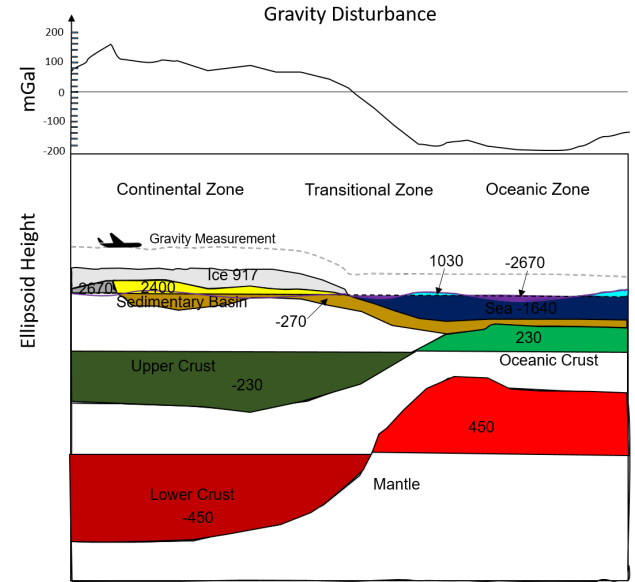
Density variation due to geology

• Free-Air Disturbance – notebook 2

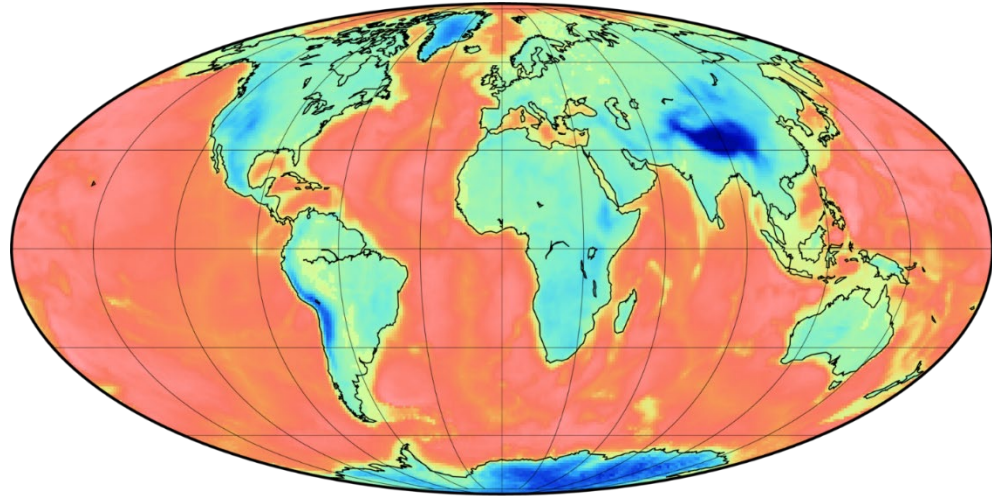
- Free-Air Disturbance



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Correct topography – notebook 3

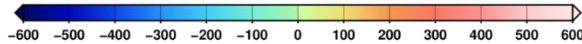


Tongji-GMMG2021S

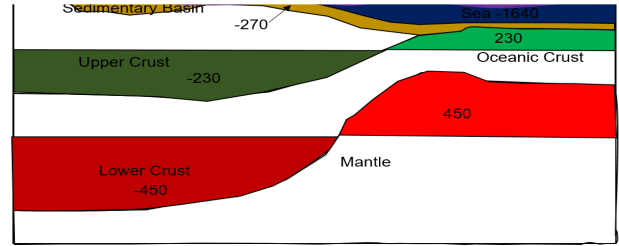
Δg_B , $1^\circ \times 1^\circ$

wrms about mean / min / max = 185.3 / -586.5 / 454.2 mgal

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Ellipsoid Height



Now we can do Inversion – notebook 4

